

Lecture 8 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 8 extends normal models from group comparisons to regression models. The key change is that each observation now has its own mean, defined as a linear function of one or more independent variables. The lecture uses age, normalized whole-brain volume, and MMSE to show how Bayesian linear models are specified, sampled, diagnosed, interpreted, and translated into effect sizes. It ends by connecting the Bayesian regression model to frequentist regression estimators, intervals, R^2 , and the F statistic.

Lecture Sections

8.1 — Bivariate data

This opening section places regression in the larger course sequence. Earlier lectures developed probability, Bayesian updating, sampling, workflow, binomial models, hierarchical models, and normal models. Linear regression now extends the same Bayesian logic to **relationships among variables**, using brain-volume data with age, MMSE, and estimated intracranial volume as motivating examples.

New Definitions and Terms

- **Bivariate data** — paired measurements on two variables for each observational unit
- **Regression model** — a model for how a dependent variable changes with one or more independent variables

Figure

Today's models relate age, brain volume, and cognitive function

- Characterize
- Estimate parameters
 - Assess effect sizes
- Make inferences

Variable	Meaning	Units
nWBV	Normalized whole-brain volume	Fraction
Age	Age at first MRI visit	years
MMSE	Cognitive screening score; higher is better	points
eTIV	Estimated total intracranial volume	cm ³

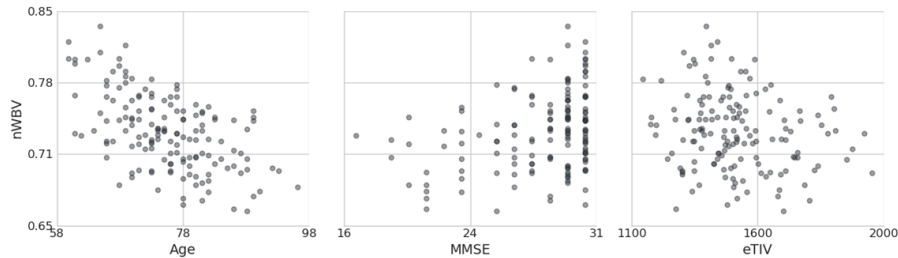


Figure 1: Slide 5 — Today's models relate age, brain volume, and cognitive function

8.2 — The bivariate model

This section builds the simple linear regression model as a direct extension of the two-group normal model. In the two-group model, each observation belonged to one of two means; in the bivariate model, each observation has its own mean. The mean is a linear function of the independent variable, while all observations share a residual standard deviation.

The model is asymmetric: x is treated as known precisely, while y is treated as measured with noise. This asymmetry may represent a plausible causal direction, as in age affecting brain volume, but it can also be a modeling convention when the causal interpretation is less clear.

New Definitions and Terms

- **Independent variable** — the predictor variable, treated as known precisely in the regression model
- **Dependent variable** — the outcome variable, modeled as noisy conditional on the predictor
- **Linear model** — a model in which the expected value of the outcome is a linear function of predictors

- **DAG** — a directed acyclic graph used to represent dependency or causal assumptions among variables

Formulas

$$\mu_i = \beta_0 + \beta_1 x_i$$

Each observation has its own expected value, determined by the regression line.

$$y_i \sim \text{Normal}(\mu_i, \sigma)$$

The observed outcome is modeled as noisy around that expected value.

$$\beta_0 \sim \text{Normal}(\mu_0, \sigma_0), \quad \beta_1 \sim \text{Normal}(\mu_1, \sigma_1), \quad \sigma \sim \text{HalfNormal}(\sigma_\sigma)$$

The priors are placed on the intercept, slope, and residual standard deviation.

Figure

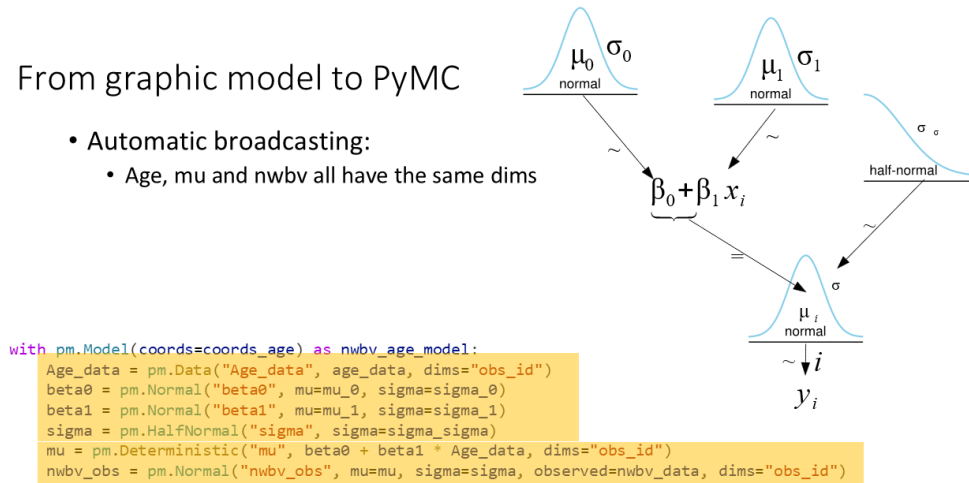


Figure 2: Slide 12 — From graphic model to PyMC

Posterior predictive shows model fit

- **Predictive data**
 - Should look like original data
 - Center and width
 - Should respect model
 - Distance from model
 - Across the range

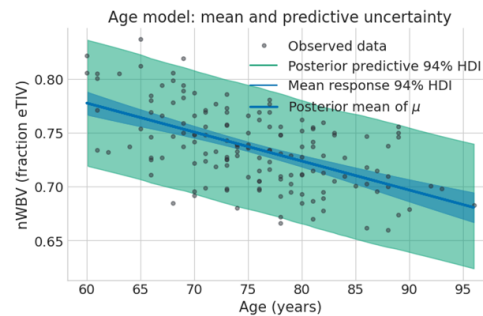


Figure 3: Slide 18 — Posterior predictive shows model fit

8.3 — Posterior regression parameters

This section interprets the posterior distribution of the simple regression model. Each posterior draw contains a possible intercept, slope, and residual standard deviation, and therefore defines a possible regression line. Uncertainty is smallest near the middle of the observed data and larger near the edges, where fewer observations constrain the line.

The main scientific quantity is the slope: in the age-nWBV model, the posterior suggests a loss of roughly 1% nWBV every 3 years. The intercept is mathematically part of the line, but because it corresponds to predicted nWBV at age 0, it is an extrapolation and is not automatically scientifically meaningful.

New Definitions and Terms

- **Posterior regression line** — a line defined by one posterior draw of the regression parameters
- **Raw effect size** — the slope expressed in the original units of the outcome and predictor
- **Standardized slope** — the slope after converting variables to standard-deviation units
- **Correlation coefficient** — in simple linear regression, the standardized slope is equal to r

Formulas

Raw slope:

$$\beta_1 = \frac{\Delta y}{\Delta x}$$

The raw effect size is the change in the dependent variable per unit change in the independent variable.

Standardized slope:

$$\beta_1^z = \frac{s_x}{s_y} \beta_1 = r$$

After standardization, the slope is dimensionless and equals the correlation coefficient in the bivariate case.

Figure

Each posterior sample is a regression line

- Each posterior sample:
 - Has β_0, β_1, σ
 - Defines a possible line
- Near middle of the data
 - More certainty
- At edges
 - Less

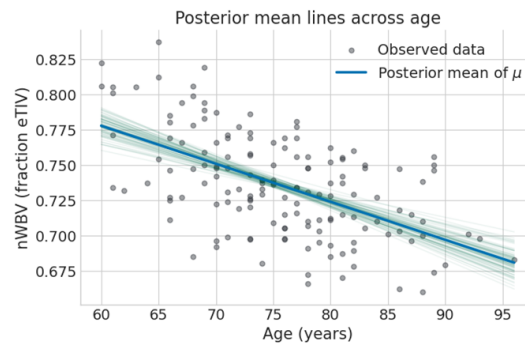


Figure 4: Slide 21 — Each posterior sample is a regression line

8.4 — Centered and uncentered data

This section shows that centering the independent variable changes the meaning of the intercept without changing the fitted line. In the uncentered model, the intercept is the expected outcome when age is 0. In the centered model, the intercept is the expected outcome when age is equal to the sample mean, which is usually more interpretable and often improves sampling.

Centering is therefore a reparameterization, not a different scientific model. The posterior for the slope and the final fit are unchanged, but the posterior for the intercept becomes narrower and more meaningful.

New Definitions and Terms

- **Centered variable** — a variable with its sample mean subtracted
- **Standardized variable** — a variable centered and divided by its sample standard deviation
- **Reparameterization** — rewriting the same model using different parameters
- **Centered intercept** — the expected outcome when the original predictor is at its mean

Formulas

Centered and standardized variables:

$$x_i^c = x_i - \bar{x}, \quad x_i^z = \frac{x_i - \bar{x}}{s_x}$$

Regression with a centered predictor:

$$\mu_i = \beta_0^c + \beta_1^c x_i^c$$

Relationship between uncentered and centered parameters:

$$\beta_1^c = \beta_1, \quad \beta_0 = \beta_0^c - \beta_1 \bar{x}$$

The slope is unchanged, while the intercept is relabeled to refer to the mean predictor value rather than zero.

Figure

Centering and standardization remove means and units

	Random variable	Data
The variable: x	$E[x] = \mu$ $\text{var}[x] = \sigma^2$	$\frac{1}{N} \sum_{i=1}^N x_i = \bar{x}$ $\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 = s^2$
Centered: x^c	$x^c = x - \mu$ $E[x^c] = 0$ $\text{var}[x^c] = \sigma^2$	$x_i^c = x_i - \bar{x}$ $\frac{1}{N} \sum_{i=1}^N x_i^c = 0$ $\frac{1}{N-1} \sum_{i=1}^N (x_i^c)^2 = s^2$
Standardized: x^z	$x^z = \frac{x - \mu}{\sigma}$ $E[x^z] = 0$ $\text{var}[x^z] = 1$	$x_i^z = \frac{x_i - \bar{x}}{s}$ $\frac{1}{N} \sum_{i=1}^N x_i^z = 0$ $\frac{1}{N-1} \sum_{i=1}^N (x_i^z)^2 = 1$

Figure 5: Slide 30 — Centering and standardization remove means and units

Intercept has different meaning in the two models

- Scientific question
 - Which do you care about?
- Statistical question
 - Which produces stable inference?
 - Because we can convert between them

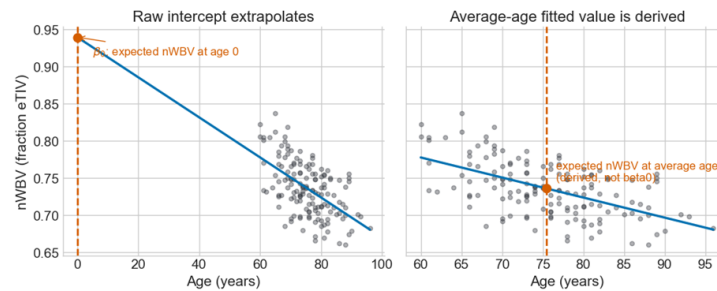


Figure 6: Slide 34 — Intercept has different meaning in the two models

8.5 — Multivariate regression

This section generalizes the bivariate model to include more than one independent variable. Age and MMSE are modeled together as predictors of nWBV, so each slope describes an effect while holding the other predictor fixed. The regression model still uses a normal likelihood, but the expected value now depends on a vector of predictors.

The lecture also emphasizes that the regression model does not assume that the independent variables themselves are normally distributed. MMSE is ordinal, bounded, and non-Gaussian, but this does not violate the normal regression likelihood because the likelihood is a model of the dependent variable conditional on the predictors.

New Definitions and Terms

- **Multivariate regression** — regression with more than one independent variable
- **Conditional effect** — the effect of one predictor when the other predictors are held fixed
- **Data matrix** — the matrix containing all predictor columns used to compute the model mean

Formulas

Multivariate regression mean:

$$\mu_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i}$$

For the lecture example, the predictors are centered age and centered MMSE.

Matrix form:

$$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$$

The data matrix $\mathbf{X}$ holds the independent variables, and $\boldsymbol{\beta}$ holds the corresponding regression coefficients.

Figure

Notice the causal assumptions

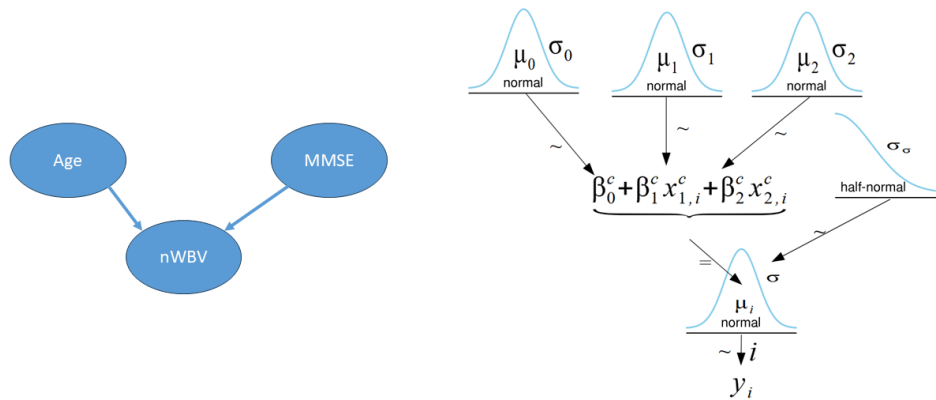


Figure 7: Slide 45 — Notice the causal assumptions

Graphical notation for a matrix model

```
with pm.Model(coords=coords) as model_matrix:
    X = pm.Data("X", X_matrix, dims=("obs_id", "var"))
    beta = pm.Normal("beta", mu=mu_beta, sigma=sigma_beta, dims="var")
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu = pm.Deterministic("mu", pm.math.dot(X, beta), dims="obs_id")
    nwbv_obs = pm.Normal(
        "nwbv_obs",
        mu=mu,
        sigma=sigma,
        observed=multiple_df["nWBV"].to_numpy(),
        dims="obs_id",
    )
```

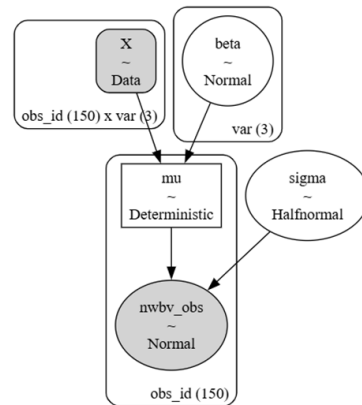


Figure 8: Slide 48 — Graphical notation for a matrix model

8.6 — Multivariate regression posterior

This section interprets the posterior of the multivariate regression. The age coefficient gives the effect of age when MMSE is held fixed, while the MMSE coefficient gives the effect of MMSE when age is held fixed. In the lecture example, nWBV decreases with age and increases with MMSE, conditional on the other variable.

Because age and MMSE have different units, the raw slopes are not directly comparable. Standardized slopes put them on a common scale, and Bayesian R^2 summarizes how much of the outcome variance is explained by the model.

New Definitions and Terms

- **Standardized conditional slope** — a multivariate regression coefficient expressed in standard-deviation units
- **Bayesian R^2** — a posterior distribution for the proportion of variance explained by the model
- **Variance explained** — the variance in the model mean relative to the total modeled variance

Formulas

Standardized slope for predictor j :

$$\beta_j^z = \frac{s_{x_j}}{s_y} \beta_j$$

This allows slopes for predictors with different units to be compared on a common scale.

Bayesian R^2 :

$$R^2 = \frac{\text{var}(\hat{y})}{\text{var}(\hat{y}) + \text{var}(e)}$$

The numerator is the variance explained by the model mean, and the denominator adds the residual variance.

Figure

Compare effect sizes using standardized slopes

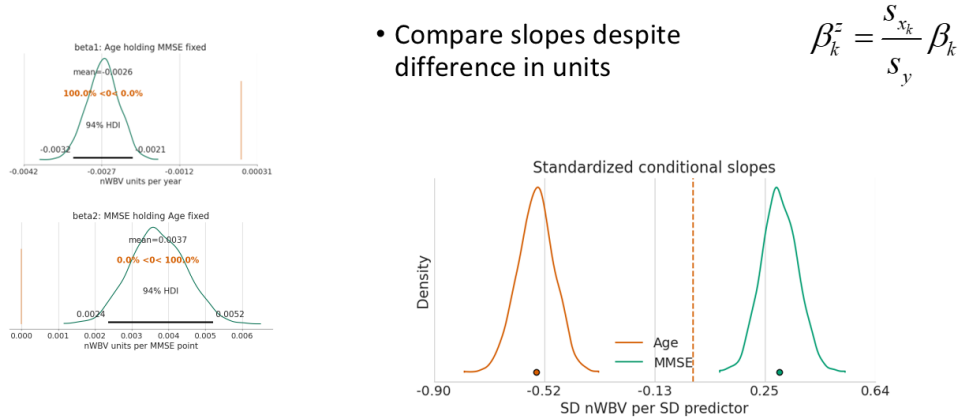


Figure 9: Slide 57 — Compare effect sizes using standardized slopes

Compare HDI and ROPE on standardized effects

- Plots show only the HDI without the tails
- No overlap with ROPE

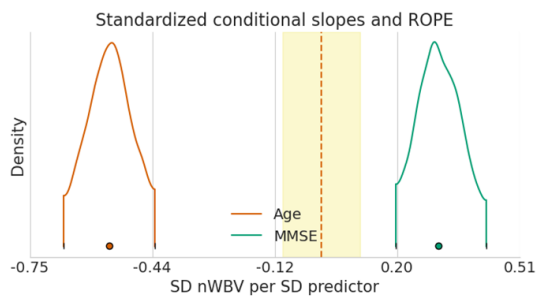


Figure 10: Slide 58 — Compare HDI and ROPE on standardized effects

Variance explained is used as an effect size

- R^2 divides the variance into two parts:
 - Variance captured by the model: $\text{var}(\mu_i)$
 - Residual variance in the data: σ^2
 - In general: $\text{var}(y) = \text{var}(\mu_i) + \sigma^2$
 - Although not exactly
- R^2 always increases when new variables are added to the model

$$R^2 = \frac{\text{var}(\mu_i)}{\text{var}(\mu_i) + \sigma^2}$$

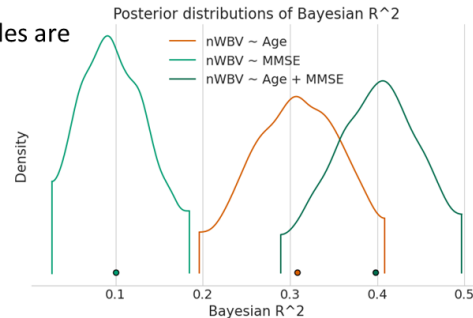


Figure 11: Slide 59 — Variance explained is used as an effect size

8.7 — Frequentist methods

This section translates the Bayesian multivariate regression model into the corresponding frequentist regression machinery. The frequentist model uses estimators, usually written with Roman letters, instead of posterior random variables written with Greek letters. The regression coefficients are estimated by minimizing the sum of squared residuals, and the sampling distribution of the estimators is used to construct intervals and hypothesis tests.

The key Bayesian contrast is that frequentist regression intervals and tests are based on sampling distributions of estimators, while the Bayesian analysis works directly with posterior distributions of parameters and derived quantities. The same model structure appears in both approaches, but the interpretation of uncertainty differs.

New Definitions and Terms

- **Estimator** — a statistic calculated from data and used as a point estimate of a parameter
- **Ordinary least squares** — the regression estimator obtained by minimizing summed squared residuals
- **Standard error** — the standard deviation of the estimator's sampling distribution
- **Regression sum of squares** — variability explained by the fitted model relative to the mean

- **Residual sum of squares** — variability not explained by the fitted model
- **Degrees of freedom** — counts used to scale sums of squares into variance estimates
- **F statistic** — a ratio of model variance to residual variance used for frequentist hypothesis tests

Formulas

Frequentist regression notation:

$$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}, \quad \hat{\mathbf{y}} = \mathbf{X}\mathbf{b}$$

The parameter vector $\boldsymbol{\beta}$ is replaced by an estimator vector $\mathbf{b}$.

Least-squares estimator:

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Residual variance estimate:

$$s^2 = \frac{(\mathbf{y} - \mathbf{X}\mathbf{b})^T (\mathbf{y} - \mathbf{X}\mathbf{b})}{N - (k + 1)}$$

Sampling distribution of the coefficient estimators:

$$\mathbf{b} \sim \text{Normal}(\boldsymbol{\beta}, \Sigma_b), \quad \Sigma_b = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

Frequentist R^2 :

$$R^2 = \frac{\text{var}(\hat{\mathbf{y}})}{\text{var}(\mathbf{y})}$$

Sums of squares:

$$SS_{reg} = (\hat{\mathbf{y}} - \bar{y})^T (\hat{\mathbf{y}} - \bar{y}), \quad SS_{res} = (\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}})$$

Degrees of freedom:

$$df_{reg} = k, \quad df_{res} = N - (k + 1)$$

F statistic:

$$F = \frac{MS_{reg}}{MS_{res}} = \frac{SS_{reg}/df_{reg}}{SS_{res}/df_{res}} \sim F(df_{reg}, df_{res})$$

Under the null model, the model variance and residual variance should be approximately equal, so the ratio should be close to 1.

Figure

The frequentist multivariate regression model

- Estimators use Roman letters instead of Greek letters
- The estimator can be derived by minimizing the sum squared error

$$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$$

$$\mathbf{y} \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$$

Estimators:

$$\hat{\mathbf{y}} = \mathbf{bX}$$

Minimize:

$$(\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}}) = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

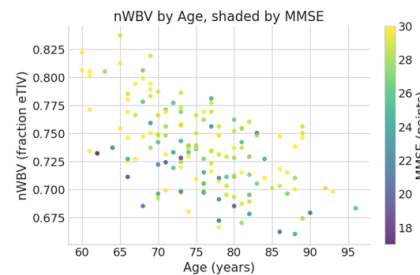


Figure 12: Slide 61 — The frequentist multivariate regression model

Compare frequentist to Bayesian intervals

$$df = N - K + 1$$

term	estimate	SE	standardized slope	t	p	95% CI lower	95% CI upper
b0	0.7361	0.0023	—	318.985	<0.0001	0.7315	0.7406
b1: Age_c	-0.0027	0.0003	-0.551	-8.675	<0.0001	-0.0033	-0.0021
b2: MMSE_c	0.0038	0.0008	0.310	4.873	<0.0001	0.0023	0.0053

Predictor	Raw effect mean	Standardized effect mean	Std. 94% HDI lower	Std. 94% HDI upper	Posterior in ROPE	HDI overlaps ROPE
Age	-0.0026	-0.547	-0.664	-0.429	0.000	no
MMSE	0.0037	0.302	0.192	0.424	0.001	no

Figure 13: Slide 65 — Compare frequentist to Bayesian intervals

Define some useful statistics

- The total variability in the model
 - The sum squared distance between the model and the mean
- The total variability not explained by the model
 - The squared distance between the data and the model

$$SS_{\text{reg}} = (\hat{\mathbf{y}} - \bar{y})^T (\hat{\mathbf{y}} - \bar{y}) = \sum_{i=1}^N (\hat{y}_i - \bar{y})^2$$

$$SS_{\text{res}} = (\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}}) = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Figure 14: Slide 67 — Define some useful statistics

8.8 — Next time

The next lecture extends regression by combining continuous predictors with categorical variables. This connects the regression framework developed here to ANOVA-style models and group comparisons.